

AdaBoost from Scratch in Python

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Decision trees ([Note 100](#)); Boosting ([Note 101](#)); Hyperparameter tuning ([Note 111](#)).
- **Compare with:** Gradient boosting ([Note 120](#)).

1. Overview

Key point: We code three stages of AdaBoost by hand on 10 observations: weights, stump, error, alpha, weight update, normalisation and upsampling, then the weighted vote. The result matches the steps of the step-by-step Note.

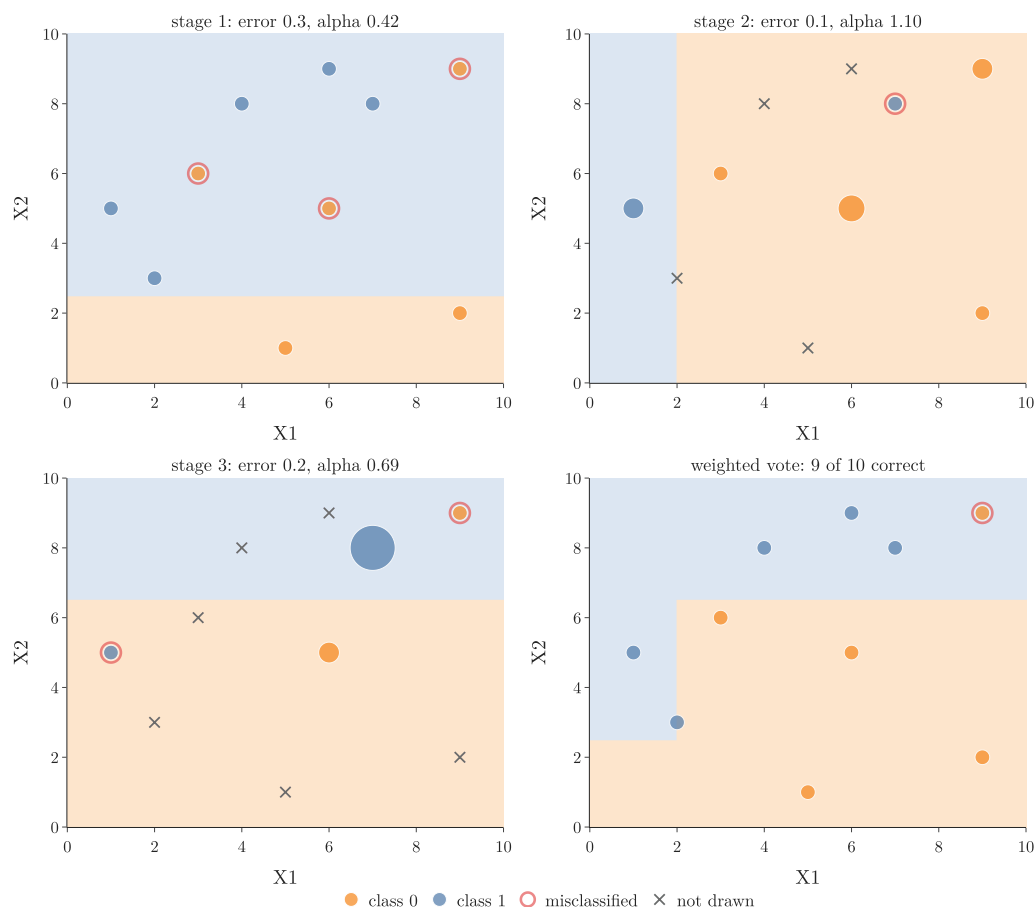


Figure 1: Three stages of AdaBoost on the 10-observation toy data, and their weighted vote. Marker size shows how many copies of a observation are in that stage's dataset; crosses are observations not drawn

The [AdaBoost step-by-step Note](#) went through one stage on paper. Here we write each step in Python with pandas, NumPy and scikit-learn, and run three stages (Figure 1). The Notebook (`notebook.ipynb`) holds the full code; `images/figs.py` draws the figures with the same data and random seed.

Along the way we see two things the paper version did not show: why the weight update uses the exponential, and what to do when a stump makes no mistakes at all.

2. The data

Key point: 10 observations, features X1 and X2, classes 1 and 0; no single straight cut separates them.

Each **observation** is one record (one row of the table). X1 and X2 are the **features**, the input variables (columns). The **target label** is the class we predict.

Row	X1	X2	label
0	1	5	1
1	2	3	1
2	3	6	0
3	4	8	1
4	5	1	0
5	6	9	1
6	6	5	0
7	7	8	1
8	9	9	0
9	9	2	0

The rows are numbered from 0, as pandas does. The classes are mixed (Figure 1, top left), so no single straight line separates them: a stump alone will make mistakes, which is exactly what we want to watch.

The labels are stored as 1 and 0, the way scikit-learn expects them. For the final vote we turn them into +1 and -1 (section 9).

Python: Building the table.

```
import pandas as pd
```

```
df = pd.DataFrame()
```

```
df["X1"] = [1, 2, 3, 4, 5, 6, 6, 7, 9, 9]
```

```
df["X2"] = [5, 3, 6, 8, 1, 9, 5, 8, 9, 2]
```

```
df["label"] = [1, 1, 0, 1, 0, 1, 0, 1, 0, 0]
```

Assigning a list to `df["X1"]` creates a column named X1.

3. Stage 1: weights, stump, error and alpha

Key point: Every observation gets 0.1; the first stump is “X2 at most 2.5 means class 0”; it misclassifies observations 2, 6 and 8, so error = 0.3 and alpha = 0.42.

Weights. With 10 observations, each observation starts at $1/10 = 0.1$.

The stump. A `DecisionTreeClassifier` with `max_depth=1` is a decision stump. Trained on all 10 observations, it chooses $X_2 \leq 2.5$: below, class 0; above, class 1 (Figure 1, top left).

Python: Weights and the first stump.

```
from sklearn.tree import DecisionTreeClassifier

df["weights"] = 1 / df.shape[0]    # 0.1 for every row
X = df[["X1", "X2"]].values
y = df["label"].values
dt1 = DecisionTreeClassifier(max_depth=1, random_state=0)
dt1.fit(X, y)
df["y_pred"] = dt1.predict(X)
df.shape[0] is the number of rows. export_text(dt1) from sklearn.tree prints the
split as text.
```

Extra: On this data the cut “ $X_1 \leq 2.5$ ” is exactly as good as “ $X_2 \leq 2.5$ ”: both get 3 observations wrong. When two cuts tie, scikit-learn tries the features in a random order, so the winner depends on `random_state` (scikit-learn docs, `DecisionTreeClassifier`). In the Notebook, seeds 0, 1, 5, 6, 7 and 8 give the X2 cut; seeds 2, 3, 4 and 9 give the X1 cut.

Mistakes. Comparing `label` with `y_pred` observation by observation, the stump is wrong on observations **2, 6 and 8**: three class-0 observations in the class-1 region.

Error and alpha. The error is the total weight of those observations, $0.1 + 0.1 + 0.1 = 0.3$. Using the formula of the step-by-step Note, section 6:

$$\alpha_1 = \frac{1}{2} \ln \left(\frac{1 - 0.3}{0.3} \right) = \frac{1}{2} \ln(2.333) = \frac{1}{2} \times 0.847 = 0.4236$$

Python: Error and alpha as functions.

```
import numpy as np

def model_error(d):
    wrong = d["label"] != d["y_pred"]
    return d.loc[wrong, "weights"].sum()

def calculate_model_weight(error, eps=1e-10):
    return 0.5 * np.log((1 - error + eps) / (error + eps))
```

```
alpha1 = calculate_model_weight(model_error(df)) # 0.4236
```

`d.loc[wrong, "weights"]` keeps only the rows where `wrong` is `True`. `np.log` is the natural log. `eps` is explained in section 7.

4. Updating the weights

Key point: The three mistakes go from 0.1 to 0.1528; the seven correct observations from 0.1 to 0.0655.

With $\alpha_1 = 0.4236$ (the step-by-step Note, section 7):

$$\text{misclassified: } 0.1 \times e^{0.4236} = 0.1 \times 1.5275 = 0.1528$$

$$\text{correct: } 0.1 \times e^{-0.4236} = 0.1 \times 0.6547 = 0.0655$$

Python: The update for every observation at once.

```
def update_row_weights(d, alpha):
    correct = d["label"] == d["y_pred"]
    return np.where(correct,
                    d["weights"] * np.exp(-alpha),
                    d["weights"] * np.exp(alpha))
```

```
df["updated_weights"] = update_row_weights(df, alpha1)
```

`np.where(condition, a, b)` takes `a` where the condition is `True` and `b` where it is `False`, row by row.

5. Why the update uses the exponential

Key point: e^α grows fast with α : a stump we trust a lot changes the weights a lot; a stump we barely trust changes them little.

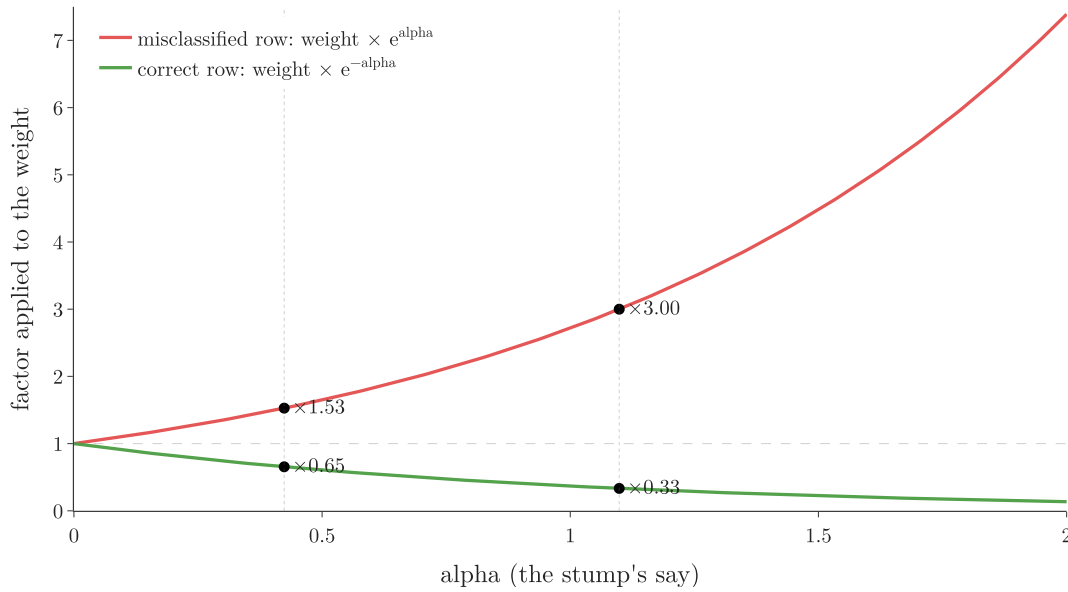


Figure 2: The factors e^α (mistakes) and $e^{-\alpha}$ (correct observations) against alpha, marked at our stage 1 and stage 2 alphas

Figure 2 plots the two factors.

- e^α (**red**) is 1 at $\alpha = 0$ and rises quickly. A large alpha means a trustworthy stump, one with few mistakes. If such a stump still gets a observation wrong, that observation is likely a hard one, and its weight is multiplied by a big number: $\times 3.00$ for our stage 2 stump ($\alpha = 1.10$).
- $e^{-\alpha}$ (**green**) is the mirror image: below 1 for any positive alpha, and smaller the larger alpha is. Observations a trustworthy stump gets right lose most of their weight: $\times 0.33$ at $\alpha = 1.10$.
- A stump we barely trust (small alpha) changes the weights only a little: $\times 1.53$ and $\times 0.65$ for our stage 1 stump ($\alpha = 0.42$).

So the size of the update follows how much we trust the stump that made it.

6. Normalising and drawing the next dataset

Key point: The updated weights add to 0.9165; after dividing, mistakes weigh 0.1667 and correct observations 0.0714. Ten random draws then give the dataset for stage 2.

Normalise. The updated weights add up to $3 \times 0.1528 + 7 \times 0.0655 = 0.9165$. Dividing each by 0.9165:

$$\text{misclassified: } \frac{0.1528}{0.9165} = 0.1667 \quad \text{correct: } \frac{0.0655}{0.9165} = 0.0714$$

Ranges. The running total of the normalised weights, `np.cumsum`, gives each observation's

upper end; its lower end is the upper end minus its own weight. Observation 0 owns 0 to 0.0714, observation 1 owns 0.0714 to 0.1429, observation 2 owns 0.1429 to 0.3095, and so on.

Upsampling. We draw 10 random numbers between 0 and 1 and, for each, pick the observation whose range it falls in (the step-by-step Note, section 9).

Python: Ranges and the new dataset.

```
df["normalized_weights"] = (df["updated_weights"]
                             / df["updated_weights"].sum())
df["cumsum_upper"] = np.cumsum(df["normalized_weights"])
```

```
rng = np.random.default_rng(0)
```

```
def create_new_dataset(d):
    indices = []
    for _ in range(d.shape[0]):
        a = rng.random()
        # first row whose upper end is above a
        upper = d["cumsum_upper"].values
        indices.append(int(np.argmax(upper > a)))
    return indices
```

```
index_values = create_new_dataset(df)
second_df = (df.iloc[index_values][["X1", "X2", "label"]]
             .reset_index(names="row"))
```

`rng.random()` returns a random number between 0 and 1; seed 0 makes the draw repeatable. `upper > a` is a list of True/False; `np.argmax` returns the position of the first True. `reset_index(names="row")` numbers the new rows 0 to 9 and keeps the original row number in a column `row`.

Our draw picks observations **6, 2, 0, 0, 8, 8, 6, 7, 6, 9**. The three mistakes (observations 2, 6 and 8) take 6 of the 10 places; observations 1, 3, 4 and 5 are not drawn at all (Figure 1, top right: crosses).

Weights start again. In the new dataset every observation gets weight 0.1 again. The upsampling has already done the reweighting: a observation drawn three times counts three times.

Extra: `np.random.default_rng(0)` is NumPy's current way to make random numbers. Older code calls `np.random.random()` with no seed, so every run draws different observations and gives different stumps. Fixing the seed makes the results repeatable. NumPy can also draw by weight in one line: `rng.choice(10, size=10, p=weights)`.

7. Stage 2, and a stump with no mistakes

Key point: The second stump is “X1 at most 2 means class 1”; it gets one observation wrong, so error = 0.1 and alpha = 1.10, a bigger say than the first stump’s.

On `second_df` we repeat every step:

1. weights 0.1 each;
2. the new stump is **X1 ≤ 2**: left, class 1; right, class 0 (Figure 1, top right);
3. it gets one observation wrong, original observation 7 (class 1 at X1 = 7), so the error is 0.1;
4. $\alpha_2 = 0.5 \times \ln(0.9/0.1) = 0.5 \times \ln 9 = 1.0986$.

The second stump has a larger say than the first (1.10 against 0.42) because it made fewer mistakes.

The update follows (Figure 2): the mistake becomes $0.1 \times e^{1.0986} = 0.3$, the nine correct observations $0.1 \times e^{-1.0986} = 0.0333$. Their sum is $0.3 + 9 \times 0.0333 = 0.6$; after normalising, the mistake weighs **0.5** and every other observation 0.0556. The draw for stage 3 gives observation 7 six times out of ten.

When a stump makes no mistakes. On a small dataset a stump can classify every observation correctly. Its error is then 0, and the formula breaks:

$$\alpha = \frac{1}{2} \ln \left(\frac{1 - 0}{0} \right) \quad \text{divides by zero}$$

The fix is to add a tiny number, such as 10^{-10} , to the error before dividing; that is the `eps` in `calculate_model_weight`. Such a stump then gets a huge but finite alpha. scikit-learn handles this case itself: it stops boosting early, because a perfect stump leaves nothing to fix (scikit-learn source).

8. Stage 3

Key point: The third stump, trained on the third dataset, is “X2 at most 6.5 means class 0”; error 0.2, alpha 0.69.

The third dataset is observations 7, 6, 7, 6, 7, 0, 7, 7, 8, 7. The new stump is **X2 ≤ 6.5**: below, class 0; above, class 1 (Figure 1, bottom left). The third stump gets two observations wrong, original observations 0 and 8, so the error is 0.2 and

$$\alpha_3 = \frac{1}{2} \ln \left(\frac{1 - 0.2}{0.2} \right) = \frac{1}{2} \ln 4 = 0.6931$$

Extra: A common slip when copying the stage code is to train the third stump on the second dataset while comparing its predictions with the labels of the third. The predictions then belong to different observations from the labels, and the “error” comes

out meaninglessly high. The Notebook makes this slip on purpose and gets an “error” of 0.7 and a negative say, $\alpha = -0.42$. Each stump must be trained and scored on the same dataset; the Notebook keeps one table per stage for that reason.

9. Prediction: the weighted vote

Key point: Convert each stump’s 1/0 to +1/-1, weight by alpha, add, take the sign. Observation (1, 5) gets +0.83: class 1. Observation (9, 9) gets +0.02: class 1, which is wrong.

We have $\alpha_1 = 0.4236$, $\alpha_2 = 1.0986$, $\alpha_3 = 0.6931$. A stump predicts 1 or 0; for the vote (the [AdaBoost intuition Note](#), section 5) we turn 1 into +1 and 0 into -1 with $2p - 1$.

Query (1, 5), true class 1. The stumps say: $X_2 = 5 > 2.5$, so +1; $X_1 = 1 \leq 2$, so +1; $X_2 = 5 \leq 6.5$, so -1.

$$0.4236 \times (+1) + 1.0986 \times (+1) + 0.6931 \times (-1) = 0.8291$$

Positive, so +1: class 1, **correct**.

Query (9, 9), true class 0. The stumps say: $X_2 = 9 > 2.5$, so +1; $X_1 = 9 > 2$, so -1; $X_2 = 9 > 6.5$, so +1.

$$0.4236 \times (+1) + 1.0986 \times (-1) + 0.6931 \times (+1) = 0.0181$$

Positive, so +1: class 1, **wrong**, though only just. On all 10 training observations the three stumps get 9 right; observation 8 is the one they miss (Figure 1, bottom right). With more stages, later stumps would keep working on it.

Python: The weighted vote.

```
def to_pm(p):          # 1 -> +1, 0 -> -1
    return 2 * p - 1
```

```
def adaboost_predict(query):
    query = np.array(query).reshape(1, 2)
    votes = [to_pm(m.predict(query)[0])
              for m in (dt1, dt2, dt3)]
    total = (alpha1 * votes[0] + alpha2 * votes[1]
              + alpha3 * votes[2])
    return np.sign(total)
```

```
adaboost_predict([1, 5])    # 1.0
```

reshape(1, 2) turns the pair into a table of one row and two columns, the shape predict expects. np.sign returns 1.0, -1.0 or 0.0.

10. Weights instead of upsampling: how scikit-learn does it

Key point: scikit-learn hands the weights to the stump with `sample_weight` instead of drawing observations, and its alpha has no factor 1/2; the decisions are the same. With the same tie-breaking, our loop reproduces `AdaBoostClassifier` exactly.

Upsampling is random: another seed draws other observations and can give other stumps. scikit-learn's `AdaBoostClassifier` avoids the randomness. scikit-learn never draws observations; it trains each stump with `fit(X, y, sample_weight=w)`, so the tree counts every observation in proportion to its weight when it compares cuts.

scikit-learn also uses a slightly different but equivalent bookkeeping, the **SAMME** algorithm (Zhu et al. 2009):

- $\alpha = \ln((1 - \text{error})/\text{error})$, twice our α (no factor 0.5);
- only the mistakes are multiplied, by e^α ; correct observations keep their weight; then all are normalised.

Both changes leave the result unchanged. Doubling every alpha doubles the vote total but never changes its sign. And multiplying the mistakes by $e^{2\alpha}$ gives the same normalised weights as multiplying the mistakes by e^α and the correct observations by $e^{-\alpha}$: only the ratio between the two groups matters.

Python: AdaBoost with sample weights, checked against scikit-learn.

```
from sklearn.ensemble import AdaBoostClassifier

def adaboost_weights(X, y, seeds):
    w = np.full(len(y), 1 / len(y))
    stumps, alphas = [], []
    for seed in seeds:
        stump = DecisionTreeClassifier(max_depth=1,
                                       random_state=seed)
        stump.fit(X, y, sample_weight=w)
        wrong = stump.predict(X) != y
        err = w[wrong].sum()
        alpha = np.log((1 - err) / err)    # SAMME
        w = w * np.exp(alpha * wrong)    # mistakes only
        w = w / w.sum()
        stumps.append(stump)
        alphas.append(alpha)
    return stumps, np.array(alphas)

abc = AdaBoostClassifier(DecisionTreeClassifier(max_depth=1),
                        n_estimators=3, random_state=0)

abc.fit(X, y)
seeds = [s.random_state for s in abc.estimators_]
stumps, alphas = adaboost_weights(X, y, seeds)
```

`alpha * wrong` is `alpha` for a mistake (`True` counts as 1) and 0 for a correct observation ($e^0 = 1$). scikit-learn gives each stump its own `random_state`; reusing them makes ties break the same way.

Both give alphas 0.8473, 1.2993 and 1.8458 (`abc.estimator_weights_`) and the same predictions. Without random draws, the three weighted stumps classify **all 10 observations** correctly. The first alpha, 0.8473, is exactly twice our hand-computed 0.4236.

11. Summary

Stage	Data	Stump	Mistakes	Error	Alpha
1	all 10 observations	$X_2 \leq 2.5$: class 0	observations 2, 6, 8	0.3	0.4236
2	drawn: 6, 2, 0, 0, 8, 8, 6, 7, 6, 9	$X_1 \leq 2$: class 1	observation 7	0.1	1.0986
3	drawn: 7, 6, 7, 6, 7, 0, 7, 7, 8, 7	$X_2 \leq 6.5$: class 0	observations 0, 8	0.2	0.6931

- Every step of a stage is a few lines of pandas and NumPy: weights, stump, error, alpha, update, normalise, draw.
- The exponential update makes trusted stumps change the weights a lot and doubtful ones a little.
- An error of 0 breaks the alpha formula; a tiny `eps` keeps it finite.
- Each stump must be trained and scored on its own stage's dataset.
- Our three upsampled stumps get 9 of 10 training observations right; scikit-learn's weighted stumps (SAMME, `sample_weight`) get all 10.

12. Sources

- scikit-learn developers. DecisionTreeClassifier, API reference, parameter `random_state`: the features are permuted at random at each split, so tied splits depend on the seed. scikit-learn.org, page `sklearn.tree.DecisionTreeClassifier`
- scikit-learn developers. Source file `sklearn/ensemble/_weight_boosting.py` (version 1.9): `AdaBoostClassifier.fit` stops when a stump's error is 0; `_boost` uses $\alpha = \ln((1 - \text{error})/\text{error}) + \ln(K - 1)$ and multiplies only the mistakes' weights. github.com/scikit-learn/scikit-learn
- Zhu, J., Zou, H., Rosset, S. and Hastie, T. (2009). Multi-class AdaBoost. *Statistics and Its Interface*, 2(3), 349–360. (The SAMME algorithm; with two classes it is AdaBoost.)

13. Key terms

Term	Meaning
<code>sample_weight</code>	The argument of <code>fit</code> that tells a scikit-learn model how much each observation counts
SAMME	The AdaBoost variant in scikit-learn: alpha without the factor 1/2, only misclassified observations reweighted; same decisions

Term	Meaning
Random seed	A number that fixes a random number generator so that a run can be repeated exactly